

Smart Weather Alert System - Project Plan

Objective

Build a location-based web application that allows users to sign up, save their location, and receive weather alerts on browser and phone.

User Module

Signup, Login, Forgot Password (Firebase Authentication).

Location Module

Get GPS location using Geolocation API or allow manual city selection. Save latitude, longitude, city, state, and country.

Dashboard

Show current weather, rain probability, next alert, temperature, humidity, wind speed, UV index, and pressure.

Settings

Enable/Disable notifications and choose alert types: Rain, Thunderstorm, High Wind, Flood, Heatwave, Fog, Cold Wave.

Alert Module

Backend periodically checks Weather API. If severe weather is detected, send push notifications.

Alert History

Store and display previous alerts.

Profile

Name, Email, City, Notification Preferences, Logout.

Database (Firestore)

Users collection and Alerts collection.

APIs

Geolocation API, Weather API, Firebase Authentication, Firestore, Firebase Cloud Messaging.

Backend

Java Spring Boot with Scheduler to check weather every few minutes and trigger notifications.

Notifications

Browser and Mobile Push Notifications using Firebase Cloud Messaging.

Responsive UI

Desktop, Tablet, Mobile, Dark Mode.

Future Features

Air Quality, Earthquake, Lightning, Cyclone, River Flood, Pollen Alerts.

Resume Highlights

Authentication, Firebase, Firestore, Java Spring Boot, REST APIs, GPS, Push Notifications, Responsive UI, Deployment.

Development Roadmap

Phase 1: Frontend UI

Phase 2: Firebase Authentication & Firestore

Phase 3: Weather API Integration

Phase 4: Java Spring Boot Backend

Phase 5: Push Notifications & Deployment

Suggested Admin Panel

Total Users, Alerts Sent, City-wise Users, Failed Notifications, Weather API Status.